



DEPARTMENT OF
ELECTRICAL AND COMPUTER ENGINEERING

ECE332

LAB 5: WAVEGUIDES

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Objective

In this lab you will explore the transmission and propagation of guided waves.

Concepts Covered

- Propagation of guided waves
- Waveguide cutoff frequency
- Transverse Electric (TE) and Transverse Magnetic (TM) modes
- Group velocity
- Resonance cavities

1 Pre-Lab Assignment

- Read Sections 8-6 through 8-11 in Ulaby[1].
- Consult the PDF instructions for the Systron/Donner 5000 Sweeper/Oscillator and HP 415E SWR Meter to familiarize yourself with their controls and indicators.
- Answer the following questions in your lab notebook.
 1. Which of the following types of transmission lines does **NOT** support the propagation of transverse-electromagnetic (TEM) waves? (include all that apply)
 - (a) Rectangular Waveguides
 - (b) Coaxial Cables
 - (c) Parallel-plate Lines
 - (d) Optical Fiber
 2. What is the dominant mode for transverse magnetic (TM) waves propagating through a waveguide?
 - (a) TM_{00}
 - (b) TM_{10}
 - (c) TM_{01}
 - (d) TM_{11}
 - (e) None of the above
 3. What is the dominant mode for transverse electric (TE) waves propagating through a waveguide?
 - (a) TE_{00}
 - (b) TE_{10}
 - (c) TE_{01}
 - (d) TE_{11}
 - (e) None of the above
 4. What type of filter does a resonant cavity implement?
 - (a) Low Pass
 - (b) High Pass
 - (c) Notch
 - (d) Bandpass
 - (e) None of the above
 5. Which instrument is used for measuring the Standing Wave Ratio (SWR) of the signal propagating through a waveguide?
 - (a) Systron/Donner 5000
 - (b) Hewlett Packard 415E
 - (c) Neither, you made those up!

2 Introduction

2.1 A Brief Review of Waveguides

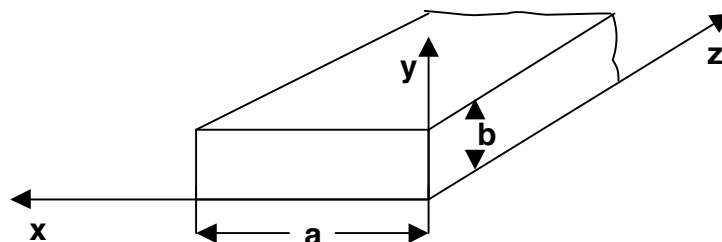


Figure 1: Coordinate system of a waveguide.

This lab examines the behavior of electromagnetic waves in a waveguide. A waveguide is a single hollow pipe (rectangular, circular, etc.) with no center conductor as illustrated in Figure 1. Hollow metallic waveguides are used as transmission lines at higher frequencies, typically ≥ 5 GHz. They have the following advantages and disadvantages:

Advantages	Disadvantages
Higher power handling capability.	Difficult to integrating them with MMIC technology.
Lower loss per unit length.	Larger dimensions than coax lines.
Simpler and lower cost mechanical structure.	Lower usable bandwidth than coax lines.

Table 1: Advantages and disadvantages of waveguides.

2.2 Rectangular Waveguides

For the case of two conductor transmission lines, the electric field of the traveling wave originates and terminates at the charged surfaces of the two respective conductors. This results in an electric field transverse to the direction of propagation and allows the electric and magnetic fields to be confined to a very small region along the axis of the transmission line. These waves are called Transverse Electro-Magnetic (TEM) waves. In a waveguide TEM waves cannot exist but different wave propagation can exist, as discussed below. We will assume that the walls of the waveguide are thicker than several skin depths. Figure 2 illustrates a parallel-plate transmission line that is analogous to a microstrip transmission line (e.g. PWB traces). Power is delivered to the load via longitudinal current flow along the upper and lower traces. Suppose that we now add a pair of short-circuited stubs of length

l . If the length of these stubs is a quarter wavelength ($\lambda/4$) then they will present an open-circuit at their input and will, therefore, not load the line with any impedance. The current on the stubs is purely reactive since no power can be absorbed, while the current along the parallel plates can deliver power to whatever load is attached at the other end. We can now

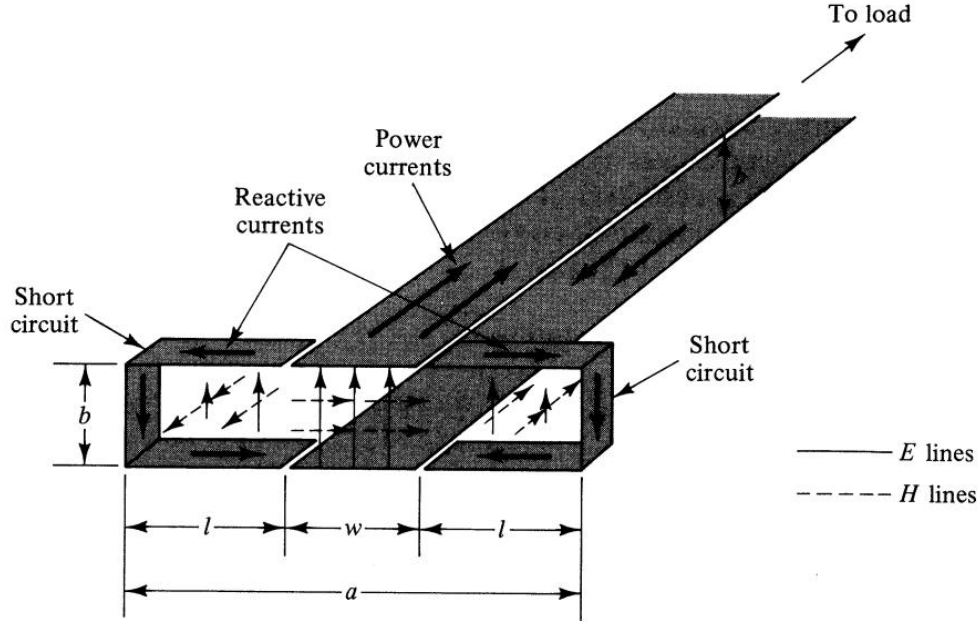


Figure 2: Rectangular waveguide as parallel-plate transmission line and $\lambda/4$ shorted stubs.

imagine that an infinite series of these short-circuited stubs is attached to the parallel-plate transmission line, effectively yielding our rectangular waveguide structure. The total width of the waveguide is $a = 2l + w$. Recall that when $l = \lambda/4$, the width w must be finite in order to support any current transport. Which for the m -th mode results in the condition,

$$a > \frac{m\lambda}{2} \Rightarrow \lambda < \frac{2a}{m}. \quad (1)$$

The frequencies therefore have to be high enough to satisfy this inequality, otherwise there will be no wave propagation down the waveguide. As the frequency decreases, shorted stub sections take up a larger portion of the total width so that less of the width is available for longitudinal conduction. The limit for the m -th mode is reached when $a = m \lambda/2$. This wavelength is the cutoff wavelength λ_c for the TE_m mode as described by,

$$\lambda_c = \frac{2a}{m}. \quad (2)$$

Field patterns inside the waveguide for the case of the TE_{10} mode are shown in Figure 3. For the TE_{mn} and TM_{mn} modes, the integer values, m and n represent the number of half

wavelengths of the traveling wave that can fit into the respective x and y dimensions of the waveguide, as shown in Figure 1, so that the cutoff frequency f_c becomes (3),(8.106)[1].

$$f_c = f_{mn} = \frac{1}{2\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \quad (3)$$

Note the existence of the waveguide wavelength λ_g which is different from the free space

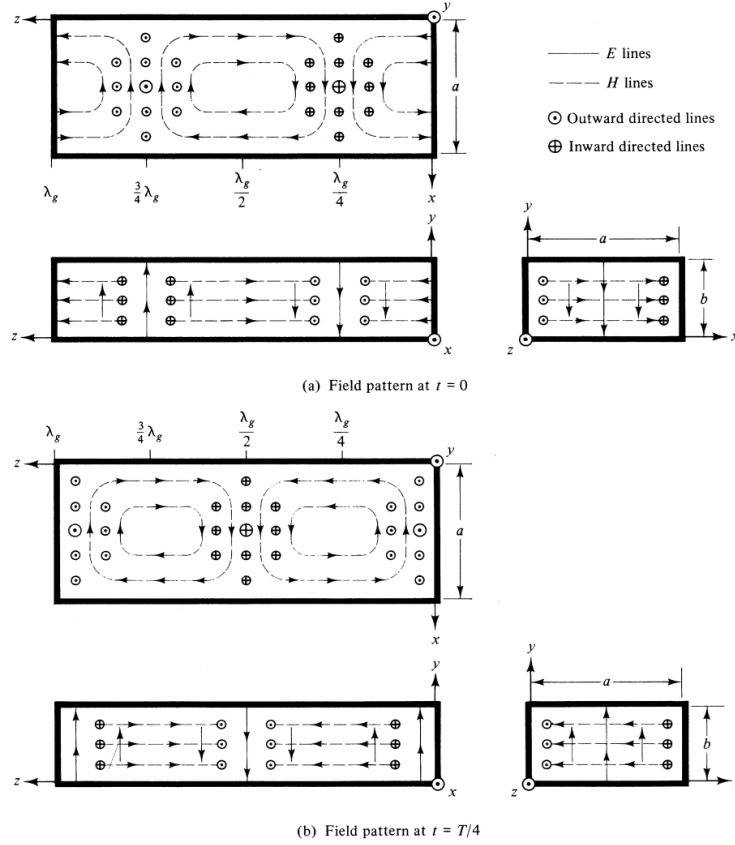


Figure 3: TE_{10} mode field patterns at different times and in different cross-sections.

wavelength and defined by,

$$\lambda_g = \frac{2\pi}{\beta} = \frac{2\pi}{\sqrt{\omega^2\mu\epsilon - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}}. \quad (4)$$

There is also a phase velocity v_p (5) associated with the propagation of the maximum of the standing wave pattern inside the waveguide.

$$v_p = f\lambda_g = \frac{c}{\lambda\sqrt{\mu_r\epsilon_r}} \frac{\lambda}{\sqrt{1 - (f_c/f)^2}} = \frac{v_{p0}}{\sqrt{1 - (f_c/f)^2}} \quad (5)$$

Notice anything strange about the equation for the phase velocity? If you look carefully, you'll see that it can be larger than the speed of light! Before you get too excited, the phase velocity is not the one that is of real interest since it does not actually transfer any information. For that we have to calculate another velocity, which is called group velocity v_g (6).

$$v_g = v_{p0} \sqrt{1 - \left(\frac{f_c}{f}\right)^2} = \frac{1}{\sqrt{\mu\epsilon}} \sqrt{1 - \left(\frac{f_c}{f}\right)^2} = \frac{1}{\sqrt{\mu\epsilon}} \frac{\lambda}{\lambda_g} \quad (6)$$

3 Experiments

Syston/Donner 5000 Oscillator and cables	Short circuit termination
SWR Meter (HP 415E)	BNC cable
Meter Stick	

Table 2: Experiments Equipment List

3.1 Wavelength Measurements

Introduction

For these experiments you will use a slotted waveguide excited by a high frequency oscillator, tunable from 8 to 12.4 GHz. A traveling probe, with a tunable crystal probe/detector, samples the electric field inside the waveguide and the signal is read on an SWR meter. The SWR meter is a high gain amplifier tuned to a frequency of around 1 kHz, so our microwave signals have to be modulated by 1 kHz signal. The SWR meter used measures the magnitude of the electric field along the slotted waveguide.

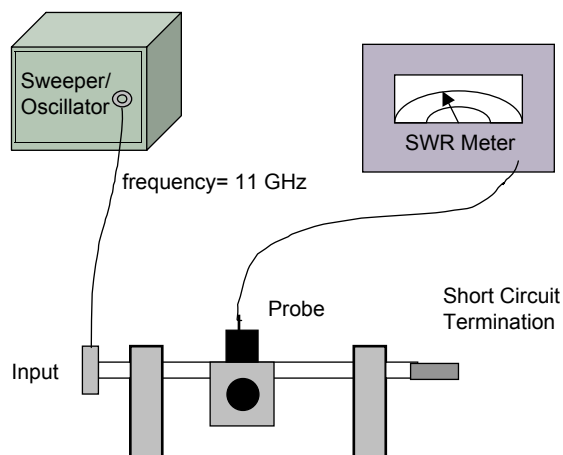


Figure 4: Schematic setup for measuring wavelength inside a waveguide.

Setup

Setup the experiment as shown in Figures 4 and 5:

- Attach the short circuit termination to the output of the waveguide.

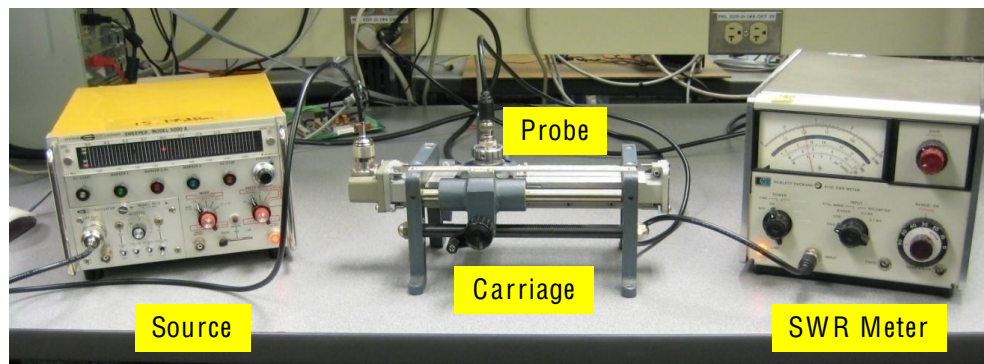


Figure 5: Experimental setup for waveguide wavelength measurement.

- Connect the probe on the top of the carriage to the SWR meter with the BNC cable.
- Power on the sweeper/oscillator by pushing the red push-button switch in lower right corner.
- Adjust the sweeper/oscillator for a frequency of 11 GHz by turning the third (from left) marker knob, labeled 2F-C (for center frequency).
- Make sure that the output amplitude of the RF Level knob is set to between 5 and 10 mW and that the “1 kHz” switch is in “ON” position (lower left corner).
- “Tune” the penetration of the probe into the waveguide. This step is necessary because a probe that is protruding too far into the waveguide will change the field profile inside the waveguide and affect the measured values. See Figure 6 for details of the carriage and probe/detector.
 - Loosen the lock that keeps the probe in place; the probe will fall all the way into the slot. This position results in a maximum signal, but also perturbs the fields inside the waveguide the most.
 - Move the probe along the carriage and find a point of maximum signal, as measured on SWR meter. You may have to adjust the “gain” setting on SWR meter so that the maximum deflection still falls within the scale.
 - Manually adjust the depth by moving the probe up and down a little bit. Penetration should not exceed the minimum required to obtain a good reading. SWR gain setting is set by the “RANGE-DB” knob. Set it to 30 or 40 (dB) and try to adjust the depth of the probe to give the deflection that is somewhere in the middle of the scale.

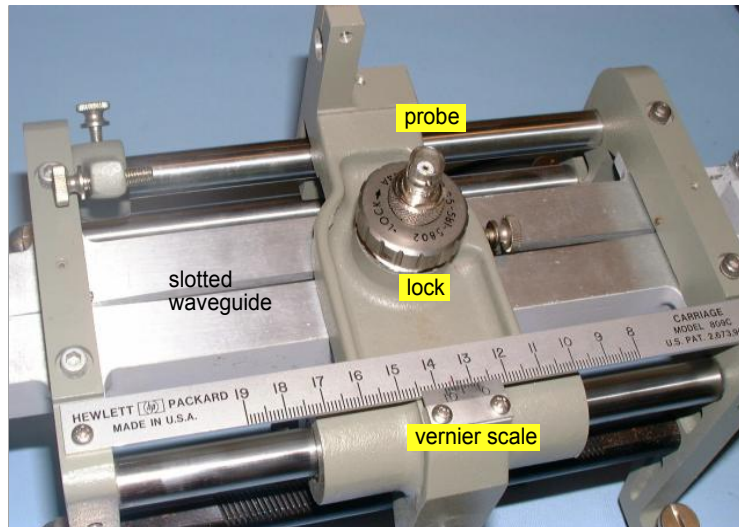


Figure 6: Details of slotted waveguide, moving probe, lock and vernier scale.

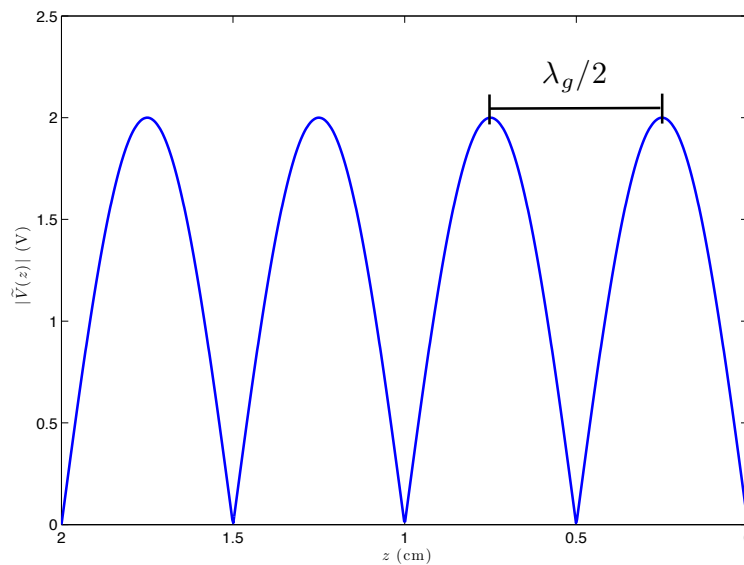


Figure 7: Expected standing wave pattern inside the waveguide.

Procedure

1. Measure the wavelength of the standing wave by adjusting the carriage to find minimum and maximum points (see Figure 6). Note the SWR measurements for the minimums are sharper than the maximums. Record three wavelength measurements, i.e. find three minima and maxima on the waveguide.
2. Repeat step 1 for input frequencies of 9.5 and 8.5 GHz. You need to normalize the SWR meter for each frequency, but it is not necessary to adjust the depth of probe tip

once it has been set.

3. Since the traveling probe cannot move to the physical end of the waveguide, measure and record the distance between the short circuit termination and the nearest detectable minimum.
4. With the oscillator turned off, measure and record the dimensions a and b of the waveguide.

Collected Data

- Measured values:
 - Three minimum and maximum standing wave values for each input frequency.
- Using (7) and the distance values in Table 3, calculate the average waveguide wavelength λ_g for each frequency.
- Using (5), calculate the average propagation velocity v_p for each frequency.
- Using (6), calculate the average group velocity v_g for each frequency.

$$\lambda_g = 4\Delta d = 4(d_{n+1} - d_n) \quad (7)$$

3.2 Frequency Measurements

Introduction

The setup is the same as Experiment 3.1, except the tunable cavity (HP X532B frequency meter) is placed between the carriage and the short circuit load, as shown in Figure 7.

Setup

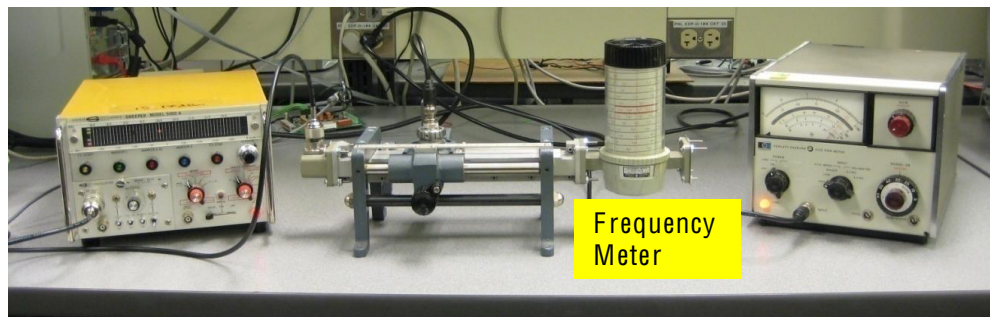


Figure 8: Setup for frequency measurement.

Setup the experiment as shown in Figure 8:

- Remove the short circuit termination from the carriage and attach it to the output of the tunable cavity (right side). Attach the tunable cavity to the output of the waveguide.
- Adjust the sweeper/oscillator for a frequency of 11 GHz by turning the third (from left) marker knob, labeled 2F-C (for center frequency).

Procedure

1. Position the probe to measure the maximum of the standing wave.
2. Measure the frequency of the wave by adjusting the resonant frequency of the tunable cavity. This is done by adjusting the black knob on top of the frequency meter.
3. At the frequency of resonance, the reading on the SWR meter will drop noticeably. Tune the resonance cavity to show the greatest drop on the SWR meter.
4. The frequency is read from the side of the tunable cavity, between the two black lines using the red line to indicate the measured value.
5. Record the cavity frequency in Table 4.
6. Repeat steps 1-5 for input frequencies of 9.5 and 8.5 GHz.

Collected Data

- Measured values:
 - Resonance frequency read from the tunable cavity for 11 GHz, 9.5 GHz and 8.5 GHz.
- Using (5), calculate the average propagation velocity v_p for each frequency.
- Using (6), calculate the average group velocity v_g for each frequency.

$f = 11$ GHz	Position (mm)	λ_g (mm)	v_p (m/s)	v_g (m/s)
Min 1				
Max 1				
Min 2				
Max 2				
Min 3				
Max 3				
Average :				
$f = 9.5$ GHz				
Min 1				
Max 1				
Min 2				
Max 2				
Min 3				
Max 3				
Average :				
$f = 8.5$ GHz				
Min 1				
Max 1				
Min 2				
Max 2				
Min 3				
Max 3				
Average :				

Table 3: Experiment 3.1 - Waveguide SWR minimums and maximums.

Oscillator f (GHz)	Measured f (GHz)	v_g (m/s)	v_p (m/s)
11			
9.5			
8.5			

Table 4: Experiment 3.2 - Resonant cavity frequencies.

Lab Report Questions

Experiment 1

Items to Turn in

- Table 3 (15 pts)

Questions

1. Explain briefly and in your own words the reason for the difference between λ and λ_g . (5 pts)
2. Since the probe measures electric field strength, what strength of signal would it detect if you could move it all the way to the short-circuit termination? In other words, what is the electric field at the shorted end of the waveguide? (5 pts)
3. Calculate the expected distance between the short circuit termination and the nearest detectable minimum. Compare the measured and calculated values. Note, that the first detectable minimum is not the same as the first minimum because the probe cannot be moved all the way to the end of the waveguide. (5 pts)
4. How many half wavelengths are in the distance from the nearest detectable minimum and the short circuit termination? (5 pts)
5. Using dimensions a and b of the waveguide that you measured:
 - (a) Calculate the cutoff frequencies of the lowest three modes using (3). (Hint: the lowest three modes may not be sequential values for m and n .) (5 pts)
 - (b) Which mode(s) are propagating? (5 pts)
 - (c) Using λ in free space, your average measured values for λ_g , and (5), recalculate the cutoff frequencies for the propagating modes. How do these calculations compare to your answers from part (a)? (5 pts)

Experiment 2

Items to Turn in

- Table 4 (15 pts)

Questions

1. Using the measured values for frequency and wavelength, calculate the phase velocity of the wave: $v_p = \lambda_g f$. How close (i.e. percentage difference) is this to your values from

Experiment 3.1? Based on this, how important is it to know the frequency accurately? (5 pts)

2. What happens to the group velocity v_g as frequency gets closer to the cutoff frequency f_c ? Can you offer some physical explanation for this? (5 pts)

References

- [1] F. Ulaby, E. Michielssen, and U. Ravaioli, *Fundamentals of Applied Electromagnetics*, 6th ed. Prentice Hall, 2010.
- [2] L. Shen and J. Kong, *Applied Electromagnetism*, 3rd ed. Cengage Learning, 2009.